TIC 94125661

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period       3.229255 d
  Epoch (t0)   1493.9737 BTJD
  Duration     0.0840 d
  Depth        25.179 %  (251,790 ppm)
  SNR          239.2

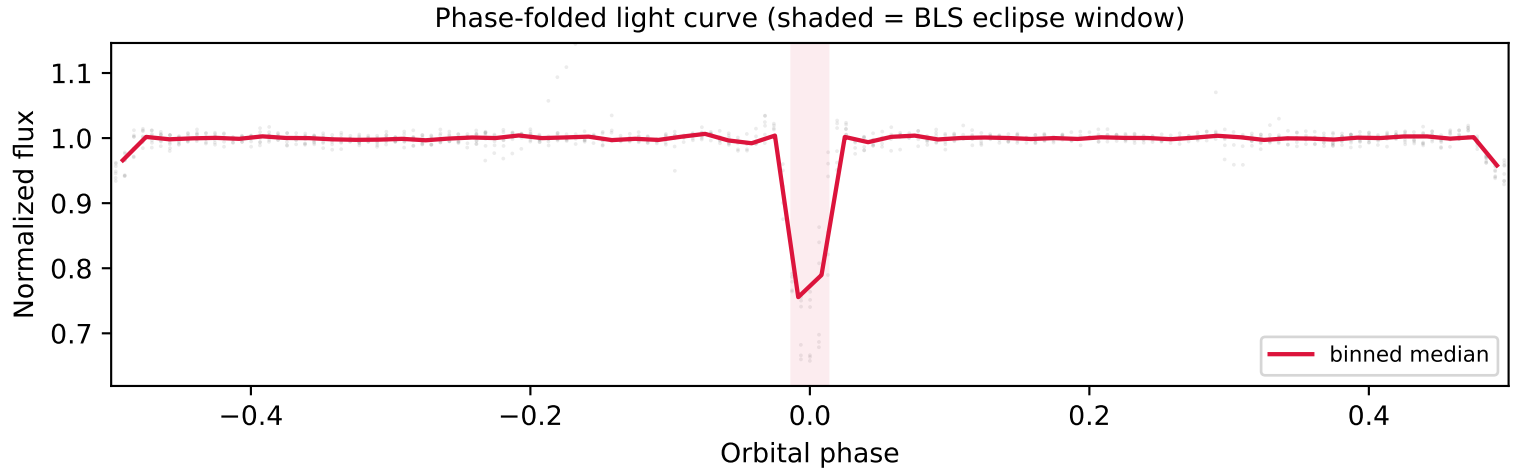
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     6  (gate >= 3)
  Robust depth ratio    0.97  (gate >= 0.50)
  Robust depth sigma    138.9  (gate >= 7.0)
  Odd/even z-score      0.20  (consistent)
  Secondary eclipse     detected, phase 0.49, 6.1 sigma
```

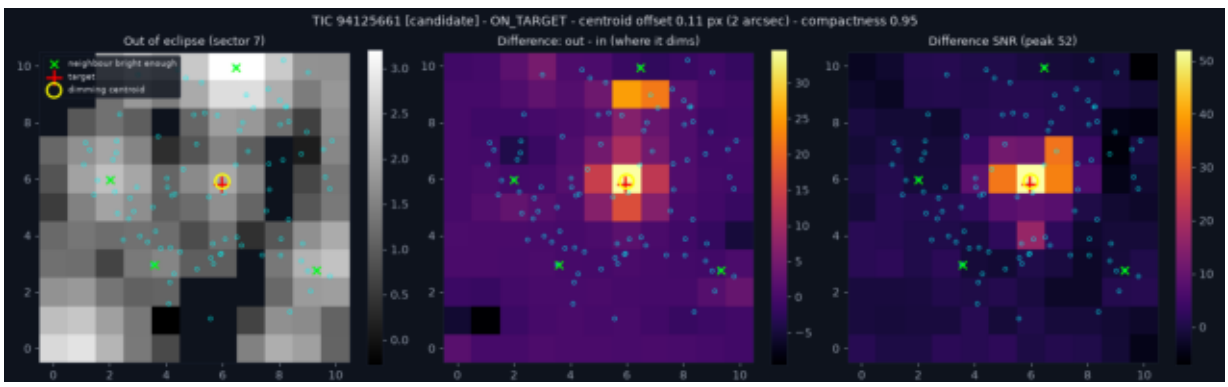
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 7    light-curve dip in this sector: 153.7 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   52.19   (gate >= 5.0)      Compactness   0.95   (gate >= 0.50)
  Dimming centroid offset     0.11 px (2 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 93959607
    3.80 px away, dTmag +1.25 - farther from the dimming than the target
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No caveats - clean detection on every check.